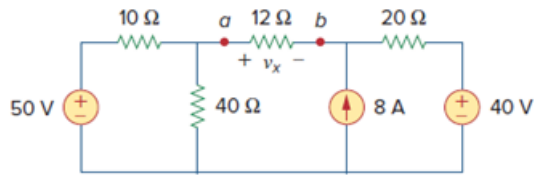


Problem

Apply source transformation to find v_x in the circuit of Fig. 4.95.

Figure. 4.95:



Step-by-step solution

Step 1 of 7

Refer to the circuit diagram depicted in Figure 4.95 in the textbook.

It is observed that, there are two series combination of voltage sources and resistors at the extreme left and right side of the circuits.

Transform the 50 V voltage source in series with $10\ \Omega$ resistor into equivalent current source in parallel with $10\ \Omega$ resistor.

Determine the equivalent current source for 50 V source.

$$I = \frac{50}{10} \\ = 5\text{ A}$$

Similarly, transform the 40 V voltage source in series with $20\ \Omega$ resistor into equivalent current source in parallel with $20\ \Omega$ resistor.

Determine the equivalent current source for 40 V source.

$$I = \frac{40}{20} \\ = 2\text{ A}$$

Step 2 of 7

Draw the modified circuit diagram after source transformations.

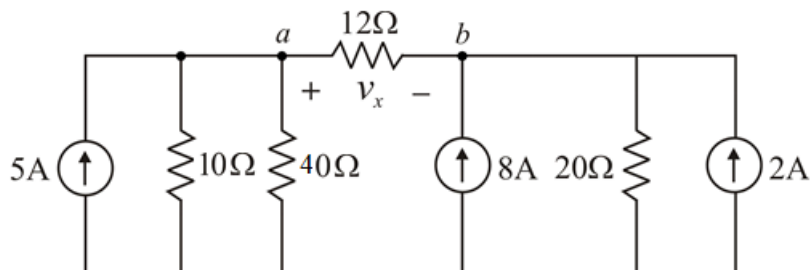


Figure 1

Step 3 of 7

In Figure 1, the resistances $10\ \Omega$ and $40\ \Omega$ are connected in parallel. Determine its equivalent resistance.

$$\Rightarrow (10) \parallel (40)$$

$$\Rightarrow \frac{(10)(40)}{10+40}$$

$$\Rightarrow 8\ \Omega$$

Draw the modified circuit diagram.

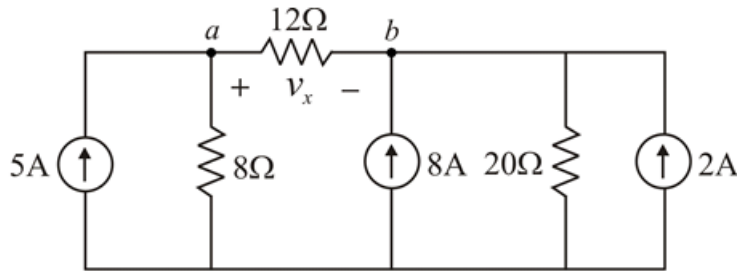


Figure 2

Step 4 of 7

In Figure 2, the current sources $8\ \text{A}$ and $2\ \text{A}$ are in parallel and have the same direction. Hence, the total current is $10\ \text{A}$. Draw the modified circuit diagram.

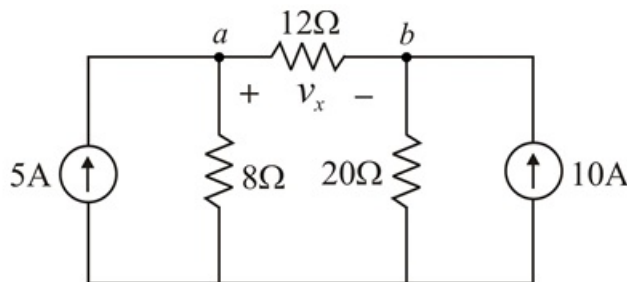


Figure 3

Step 5 of 7

In Figure 3, it is observed that, there are two parallel combination of current sources and resistors at the extreme left and right side of the circuits.

Transform $5\ \text{A}$ current source in parallel with $8\ \Omega$ resistor into equivalent voltage source in series with $8\ \Omega$ resistor. Calculate the equivalent voltage source for $5\ \text{A}$ current source.

$$\begin{aligned} V &= (5)(8) \\ &= 40\ \text{V} \end{aligned}$$

Similarly, transform $10\ \text{A}$ current source in parallel with $20\ \Omega$ resistor into equivalent voltage source in series with $20\ \Omega$ resistor. Calculate the equivalent voltage source for $10\ \text{A}$ current source.

$$\begin{aligned} V &= (10)(20) \\ &= 200\ \text{V} \end{aligned}$$

Step 6 of 7

Draw the modified circuit diagram after source transformations.

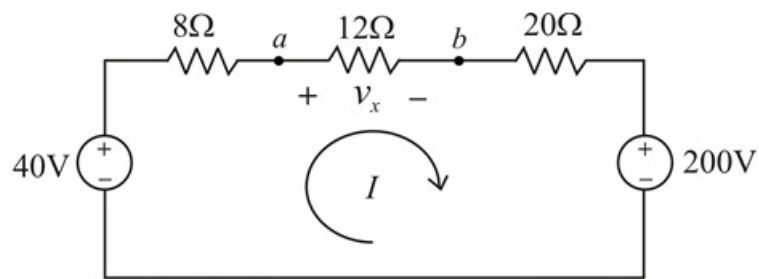


Figure 4

Step 7 of 7

Apply Kirchhoff's voltage law to the circuit shown in Figure 4.

$$-40 + 8I + 12I + 20I + 200 = 0$$

$$40I = -160$$

$$I = -\frac{160}{40}$$

$$I = -4 \text{ A}$$

Calculate the voltage v_x .

$$\begin{aligned} v_x &= 12I \\ &= 12(-4) \\ &= -48 \text{ V} \end{aligned}$$

Therefore, the voltage v_x in the circuit is -48 V.